Title: UART Communication with Letter Case Conversion and LED Color Indication

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Introduction

Before this lab, it was understood that UART (Universal Asynchronous Receiver-Transmitter) is a widely used serial communication protocol for exchanging data between microcontrollers and external devices. This lab focused on implementing an interactive UART system using the AHL-STM32L431 embedded board and the AHL-GEC-IDE. The purpose was to configure UART to convert uppercase letters (A–Z) to lowercase (a–z) and echo them back to the serial terminal while providing visual feedback via an onboard LED that changes color upon data reception.

Materials

Expendables	Apparatus
AHL-STM32L431 Board	USB Type-C Cable
Jumper Wires	Personal Computer
AHL-GEC-IDE v4.53	Serial Monitor (PuTTY/Tera Term)

Procedure

The procedure was followed as outlined in the lab handout. The key steps included:

1. Creating a new UART project in AHL-GEC-IDE and configuring the USART2 peripheral.
2. Setting up TX (PA2) and RX (PA3) pins in alternate function mode.
3. Initializing UART communication at 115200 bps.
4. Implementing a UART interrupt handler to process received data.
5. Adding logic to detect uppercase letters, convert them to lowercase, and echo the result.
6. Integrating LED control to switch the LED color to blue upon data reception and revert to red when idle.
7. Building, uploading, and testing the firmware using a serial terminal.

Core modification inside USART2_IRQHandler() in isr.c:

```
  ------------------------------------------------------------ */
void USART2_IRQHandler(void)
{
    uint8_t ch, flag;

    DISABLE_INTERRUPTS;                 // critical section

    ch = uart_rel(UART_User, &flag);      // read & clear IRQ

    if (flag)                            // real byte
    {
        /* 1. Upper ? lower */
        if (ch >= 'A' && ch <= 'Z')
            ch += 32;

        /* 2. Echo */
        uart_send1(UART_User, ch);

        /* 3. Switch LED once */
        if (mLedPin == LIGHT_RED)
}
```

Data

Observed Output (Serial Terminal):



Input: W → Output: w

Input: H → Output: h

Input: Y → Output: y

Input: I → Output: i

Input: S → Output: s

Input: S → Output: s

Input: H → Output: h

Input: E → Output: e

Input: S → Output: s

Input: O → Output: o

Input: P → Output: p

Input: R → Output: r

Input: E → Output: e

Input: T → Output: t

Input: T → Output: t

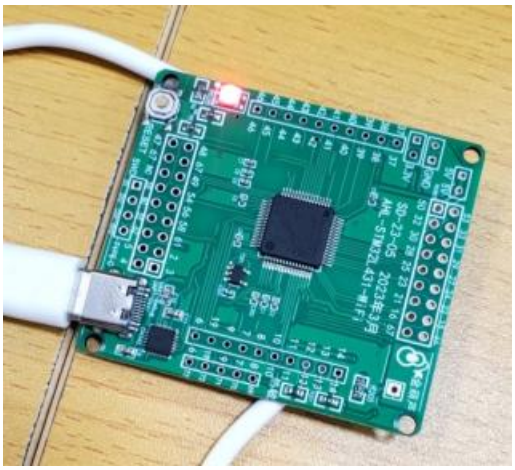
Input: Y → Output: y

Observed LED Behavior:

- LED turns **blue** when any character is received.



- LED turns **red** (default state) when idle.



UART Settings:

- **Baud rate:** 115200 bps
- **Data bits:** 8
- **Stop bits:** 1
- **Parity:** None

Variables Used:

- ch: Stores received character
- flag: Indicates if UART receive interrupt occurred
- LIGHT_RED, LIGHT_BLUE: GPIO LED macros
- uart_send1() and uart_re1(): HAL functions for UART communication

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Calculations

No mathematical computations were performed, but ASCII value manipulation was used for letter case conversion:

$$\text{Lowercase} = \text{Uppercase} + 32$$

For example, converting 'A' (ASCII 65) to 'a' (ASCII 97) by adding 32.

Analysis

The experiment successfully demonstrated bidirectional UART communication with real-time character processing and visual feedback. Key insights include:

- UART interrupts enable efficient, event-driven communication.
- ASCII arithmetic provides a simple method for text manipulation in embedded systems.
- GPIO-based LED control enhances system feedback and debugging capabilities.
  Potential errors could arise from baud rate mismatches or incorrect LED initialization.

Conclusion

The purpose of this lab was to implement a UART system that converts uppercase letters to lowercase and provides LED-based visual feedback. This was successfully achieved: all uppercase inputs were converted and echoed back correctly, and the LED color changed to blue upon data reception.

The experiment reinforced practical skills in UART interrupt handling, ASCII manipulation, and peripheral control on STM32 microcontrollers. These techniques are foundational for applications in IoT, embedded user interfaces, and serial communication diagnostics.